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Renewable energy communities and energy poverty: A novel benefit-sharing model based on differentiated pricing and incentive allocation

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ABSTRACT

The transition to sustainable energy systems requires increasing attention to the challenge of energy poverty. In this context, Renewable Energy Communities (RECs) are emerging as promising tools for promoting both decarbonization and social equity. This study aims to develop and test a new benefit-sharing model within RECs, designed to support vulnerable consumers through differentiated pricing mechanisms and targeted allocation of incentives. The methodology is based on a discounted cash flow analysis implemented in dedicated Python-based software (S-REC), which simulates energy, economic, and social flows within a REC configuration that includes prosumers and consumers with varying levels of vulnerability. The results show that the proposed benefit-sharing model improves both economic performance and social equity. Across the scenarios analyzed, the Net Present Value per unit of installed capacity (NPV/Size) ranges from 2680 €/kW to 5064 €/kW, the Profitability Index (PI) increases from 2.22 to 3.30, the Internal Rate of Return (IRR) varies between 32 % and 70 %, and the Discounted Payback Time (DPBT) decreases from 5 to 2 years. In addition, vulnerable households reduce their annual energy expenditure below the 10 % energy poverty threshold through differentiated tariffs and targeted incentive transfers. The model also demonstrates that tailored redistribution mechanisms are more effective than uniform allocation approaches in directing support to households experiencing energy poverty. These findings suggest that RECs can serve as effective energy welfare tools, combining economic efficiency with social inclusion and contributing to a more just and resilient energy transition.

Abbreviations

A_{cell}	Active module area
ARERA	Autorità di Regolazione per Energia Reti e Ambiente
BD_i	Benefit of the i-th consumer as a percentage
BD_j	Benefit of the jth prosumer as a percentage
B_i	Benefits obtained by the i-th consumer
B_j	Total revenue for the jth prosumer
B_{tot}	Sum of benefits obtained by prosumers and consumers
B_v	Benefits received by the vth vulnerable consumer
C_{ae}	Electrical connection cost
CI	Incoming cash flows
$C_{i,t}$	Inverter replacement cost at time t
$C_{inv,\ unit}$	Initial unit investment
$C_{lcs,t}$	Loan capital share cost at time t
CO	Cash outflows
DCI	Discounted cash inflows
DCO	Discounted cash outflows
DPBT	Discounted Payback Time
dE_f	Power derating
E	Equity ratio
$E_{C,\ i,r}$	Remaining energy demand from the i-th consumer
$E_{C,\ i}$	Energy required by the i-th consumer
$E_{C,j\rightarrow i}$	Energy supplied by the j-th prosumer to the i-th consumer
$E_{C,j\rightarrow v}$	Energy supplied by the jth prosumer to the vth consumer
$E_{C,v}$	Energy requirement of the vth vulnerable consumer
$E_{Ex,\ t}$	Energy exchanged with consumers at time t
$E_{Ex,i}$	Energy supplied by prosumers to the i-th consumer

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$E_{Ex,v}$	Energy supplied by prosumers to the vth consumer
$E_{Exi,\ t}$	Energy exchanged with the individual consumer at time t
$E_{Exv,t}$	Energy exchanged with the individual vulnerable consumer at time t
E_{Gj}	Energy fed into the grid by the jth prosumer
E_j	Energy available to the individual prosumer
$E_{NSCj,r}$	Remaining energy for exchange from the j-th prosumer
E_{NSCj}	Energy not self-consumed by the jth prosumer
$E_{Out,t}$	Energy produced in the first year
$E_{req,\ i}$	Energy required by an individual consumer
$E_{req,v}$	Energy required by the individual vulnerable consumer
E_{req}	Total energy demand from consumers
E_{SCj}	Energy self-consumed by the jth prosumer
GSE	Gestore dei Servizi Energetici
$G_{v,t}$	Purchases from the network by an individual vulnerable consumer at time t
G_v	Purchases from the grid by the vth vulnerable consumer
i	Consumer index
inf	Euro inflation rate
inf_{el}	Electricity inflation rate
IRR	Internal Rate of Return
j	Prosumer index
k_f	Optimal tilt angle
M	Number of consumers
N	Lifespan of the PV system
N*	Inverter replacement
N_{debt}	Payback period
NPV	Net Present Value
NPV_j	Net Present Value for the individual prosumer
NPV/Size	Net Present Value per unit of installed capacity
$n_{RSC,a}$	Number of prosumers still available for exchange
n_{RSC}	Number of prosumers
N_{TaxD}	Tax deduction period
N_v	Number of prosumers trading with the vth consumer
P	Number of vulnerable consumers
P_{Cass}	Percentage insurance cost
P_{Ci}	Percentage inverter cost
P_{Cm}	Percentage maintenance cost
P_{Ctax}	Taxes
p_t^c	Purchase price at time t
p_{t+1}^c	Purchase price at time t+1
$p_{t,v}^c$	Purchase price for the vulnerable individual consumer at time t
PI	Profitability Index
p_i^{REC}	Price for the i-th consumer in the REC
$p_{v,max}^{REC}$	Price applicable to the vth vulnerable consumer
p_0^s	Sale price at time 0
p_t^s	Sale price at time t
p_{t+1}^s	Sale price at time t+1
r	Opportunity cost of capital
$R_{C,\ j}$	Revenue from sales to consumers for the jth prosumer
r_d	Mortgage interest rate
$R_{Ex,t}$	Revenue from trading with consumers at time t
R_{salej}	Revenue from sales to the grid for the jth prosumer
R_v	Income of the vth vulnerable consumer
S	Size of the PV system
S_{billj}	Bill savings for the jth prosumer
$S_{j\rightarrow v}$	Incentives transferred by the jth prosumer to vulnerable consumers
$Sl_{el,t}$	Energy sales to consumers at time t
$Sp_{el,t}$	Energy sales at time t
$S_{REC}^u + C_{val,u}$	Premium rate + ARERA fee
$Sr_{el,t}$	Energy sales to the GSE at time t
S_{SCj}	Incentives for energy self-consumed by the jth prosumer
$S_{V,t}$	Incentives transferred to vulnerable consumers at time t
S_v	Incentives to be transferred to the vth vulnerable consumer
t	Year index
$TaxD_u$	Tax deduction
t_r	Average annual insolation
v	Index of vulnerable consumers
VAT	Value added tax
$V_{\ t}$	Incentives required for vulnerable consumers at time t
V_v	Incentives required for the vth consumer based on their expenditure
η_{bos}	System efficiency balance
η_m	Module efficiency
$\overline{\omega}_{self,c}$	Average self-consumption
$\omega_{self,j}$	Self-consumption of the individual prosumer

1. Introduction

The energy transition is one of the fundamental pillars of sustainable development, as it intertwines environmental, economic and social objectives within a single trajectory of systemic transformation. The decarbonization of energy systems, the spread of renewable sources and the improvement of energy efficiency are not only tools for mitigating climate change, but also levers for promoting social equity, territorial resilience and collective well-being [1–3]. Energy poverty, defined as the difficulty in accessing adequate energy services, is a global challenge with significant social, economic, and environmental implications, with important consequences for human health [4]. Although direct support measures such as social tariffs can provide temporary relief, they do not address the structural causes of this condition [5]. In this context, energy is increasingly recognized as a fundamental right, as access to adequate energy services is closely linked to health, social inclusion and opportunities for human development [4,6,7].

However, the persistence of energy poverty continues to pose a structural challenge for many socio-economic contexts, both in developing countries and in advanced economies [8–10]. This highlights the need for integrated approaches that combine technological, institutional and social dimensions, geared towards long-term solutions and consistent with the Sustainable Development Goals [11,12].

This is the context for Renewable Energy Communities (RECs), participatory models of local renewable energy production and consumption based on resource sharing, democratic governance and the redistribution of economic and environmental benefits [13,14]. Their development has been strongly supported by the European Union through the Renewable Energy Directive (RED), which formally introduced RECs under RED II (Directive (EU) 2018/2001) and further strengthened their role under RED III (Directive (EU) 2023/2413). In line with these directives, RECs are recognize RECs as key instruments for accelerating the deployment of renewable energy, fostering citizen participation, promoting social inclusion, and strengthening territorial cohesion [15,16].

Despite growing recognition of their potential, literature still lacks software-based decision-support tools capable of integrating energy, economic, and social dimensions within a single analytical framework. In particular, few studies provide operational models that explicitly incorporate differentiated pricing and targeted incentive redistribution to evaluate ex ante the impact of RECs on vulnerable households and energy poverty reduction. This gap is particularly critical in contexts characterized by high socio-economic vulnerability, where robust ex ante assessments are essential to guide policies and investments toward equitable and sustainable outcomes [17–19].

In light of these considerations, there is a need to integrate the social dimension, particularly the fight against energy poverty, into decision-support tools capable of operating in a dynamic and territorial manner. Such models can bridge the gap between strategic vision and operational implementation, helping to make energy communities not

only technological solutions for the green transition, but also concrete levers for a just and inclusive transition.

This study aims to develop and validate a new benefit-sharing model for RECs that integrates economic efficiency with social equity, with a particular focus on tackling energy poverty. Vulnerable households are defined as those experiencing energy poverty, identified as cases in which annual energy expenditure exceeds 10 % of household income. They are therefore considered vulnerable in terms of both energy access and income constraints. The proposed model combines differentiated pricing and targeted incentive allocation to reduce energy costs for these households and ensure that their energy expenditure does not exceed this threshold, while maintaining the economic sustainability of prosumers. To evaluate its effectiveness, a simulation-based approach is implemented through a dedicated software tool that analyzes energy, economic, and social flows within different REC configurations. The objective is to assess whether the model can simultaneously improve overall profitability, promote a more equitable distribution of benefits among participants, and contribute to the reduction of energy poverty.

The paper is organized as follows. Following this introduction, Section 2 presents a literature review aimed at framing the research topic. Section 3 describes the methodology adopted, including the economic model, input data, and the development of the Python software tool named S-REC (Simulator of Renewable Energy Community). Section 4 presents the results, focusing on the economic indicators of a 20 kW photovoltaic system in a REC configuration, consisting of four prosumers and several consumers - including users in energy-vulnerable situations - while also analyzing the mechanisms for distributing benefits. Finally, Section 5 discusses the results obtained, and Section 6 presents the main conclusions of the study.

2. Literature review

The literature on RECs has developed along three main axes: their role in energy transition, their socio-economic dimension, and institutional governance. Numerous studies recognize RECs as key players in the spread of renewable sources and the reduction of emissions, thanks to their ability to promote distributed production and collective self-consumption, with both environmental and economic benefits [20–22].

A significant strand of research focuses on the motivations for prosumer participation, highlighting how economic savings are a key incentive, alongside social and environmental factors such as local solidarity, environmental protection and a stronger sense of community belonging [23–25]. From this perspective, RECs are interpreted not only as technical energy management tools, but also as social institutions capable of promoting new forms of energy citizenship and democratic participation [26,27].

A large part of the literature analyses the contribution of RECs to reduce energy poverty. European and international studies show that sharing locally produced energy, particularly from photovoltaic (PV) systems, reduces supply costs and improves access to energy services for the most vulnerable groups [28,29]. This approach is often associated with the concept of energy justice, understood as the equitable distribution of the benefits and burdens of the energy transition [30].

The long-term sustainability of RECs is strongly influenced by governance models and regulatory frameworks. The literature highlights how transparency, regulatory stability and the inclusiveness of public policies are essential conditions for reducing technical, legal and economic barriers to participation, particularly for low-income households [31,32]. At the same time, the need for decision-making structures that balance the individual interests of prosumers with the collective goals of the community is emphasised in order to ensure equity and social cohesion [33].

A further area of research concerns the redistribution of the economic benefits generated by RECs. Several studies propose models for the allocation of revenues and incentives that take into account both economic efficiency and social impact, demonstrating that energy communities can be more cost-effective than individual self-consumption programmes, especially when geared towards supporting vulnerable households [34,35].

The territorial dimension is another central issue. Empirical evidence shows that the effectiveness of RECs varies according to geographical, socio-economic and institutional contexts, with significant differences between urban and rural areas and between different European countries [36–38]. This reinforces the idea that there is no uniform path to decarbonisation, but that strategies adapted to local specificities and networks of actors in the territories are necessary [39–41].

More recent studies also highlight the role of digitalisation and advanced analytical tools in improving operational efficiency and collective learning capacity within RECs [42,43]. In this regard, some contributions propose the use of decision-making models and software platforms to simulate energy and economic scenarios, assess the profitability of investments and estimate social impacts, particularly in the contexts of public construction and social housing [44].

Overall, the literature recognizes RECs as key instruments for a just energy transition, capable of integrating environmental sustainability, social equity, and local development objectives. However, few studies provide operational, software-based decision-support tools that simultaneously integrate energy, economic, and social dimensions within a single analytical framework. In particular, limited attention has been devoted to models that explicitly incorporate differentiated pricing and targeted incentive redistribution to assess ex ante the impact of RECs on vulnerable households and energy poverty reduction. This study addresses this gap by developing a simulation-based model and dedicated software tool to evaluate the economic feasibility of RECs and the distribution of benefits among prosumers and consumers, including households experiencing energy poverty [17,45].

3. Methodology

The financial feasibility analysis of a REC project based on PV systems, which also involves vulnerable consumers, was conducted using the Discounted Cash Flow (DCF) method, widely used in the literature to assess the viability of investments in the energy sector [22,46–48]. The model was appropriately adapted to account for the presence of vulnerable consumers and the transfer of a portion of the incentives derived from shared energy to households experiencing energy poverty. Based on this approach, the main financial indicators, as NPV, NPV/Size, PI, IRR and DPBT, were calculated, both in the absence and in the presence of state support.

3.1. Economic model

This study builds on a sequence of previous contributions on the economic assessment and benefit-sharing mechanisms of RECs [11,15,18,22]. These studies progressively developed the economic evaluation of PV-based RECs, the allocation of benefits among prosumers and consumers, and the integration of social-equity criteria. Table 1 summarizes how the present study extends these previous approaches.

Compared with these previous studies, the present work introduces a more operational and socially targeted approach. Rather than only redistributing benefits among prosumers or stakeholder categories, it explicitly models vulnerable consumers within the REC and adjusts both internal energy prices and incentive transfers to ensure that their annual energy expenditure does not exceed 10 % of household income.

Analyzing the perspective of prosumers, who effectively bear the costs associated with the installation of PV systems, there are four sources of revenue. Firstly, the annual savings on bills achieved by self-consuming the renewable energy produced, replacing the energy supplied by the grid at market price. Excess energy can be traded locally with consumers within the community, at an appropriate price derived from their economic circumstances, lower than the respective market purchase price and higher than the price at which it is sold to the grid. In

Table 1
Literature Background and Novel Contributions of the Present Study.

Study	Main contribution	Why it is relevant	Limitation addressed in this study
[22]	Provides an economic assessment of PV-based RECs under different market and incentive scenarios and compares alternative benefit-sharing rules among prosumers and an additional consumer.	It represents the starting point for modelling REC profitability and benefit allocation based on self-consumption and local energy exchange.	It does not explicitly model vulnerable consumers or targeted mechanisms for reducing energy poverty.
[11]	Extends REC benefit-sharing by introducing profit distribution models that reward virtuous behaviour and include income-based criteria.	It is relevant because it links economic performance with fairness criteria among renewable self-consumers.	Vulnerability is treated mainly through income-based redistribution among prosumers, without a dedicated mechanism for vulnerable consumers purchasing energy from the REC.
[15]	Provides socioeconomic evidence on citizens' willingness to participate in RECs and to redistribute part of their benefits toward households in need and collective purposes.	It supports the social acceptability of redistribution mechanisms within RECs.	It focuses on social preferences and general redistribution attitudes, rather than on an operational economic model for vulnerable consumers.
[18]	Proposes a broader stakeholder-based benefit distribution model, including producers, consumers, households in economic difficulty, territorial redevelopment, the State, and ESCOs.	It is relevant because it broadens REC benefit allocation beyond prosumers and introduces explicit stakeholder categories linked to social equity.	It does not define a consumer-level pricing and incentive-transfer mechanism ensuring that vulnerable households fall below the 10 % energy-poverty threshold.
Present study	Develops a software-based REC model combining differentiated tariffs and targeted incentive transfers for vulnerable consumers.	It integrates economic feasibility, local energy exchange, benefit distribution, and energy-poverty mitigation in a single operational framework.	It advances previous models by explicitly identifying vulnerable households as those whose energy expenditure exceeds 10 % of income and by calibrating prices and incentive transfers to reduce this burden below the threshold.

the case of vulnerable consumers only, the exchange tariff may coincide with the latter, the lowest possible price. The remaining energy is fed into the national grid and sold at the selling price. Furthermore, incentives are available for self-consumed energy and for energy shared locally with consumers. It should be noted that, in the Italian context, state subsidies consist of two components: an incentive tariff, which depends on the size of the plant, its location and the hourly zonal price, and the ARERA valuation component, which stood at around 8 €/MWh in 2025 and varies annually according to the Authority's decisions. Revenues from the incentive scheme are calculated as the product of the percentage of self-consumed energy, the energy produced annually by the PV systems and the amount of the state subsidies defined above; furthermore, where consumers are present, the product of the annually shared energy (equal to the minimum between the energy required by consumers and the difference between the energy produced each year and the share of self-consumed energy) and the incentives for which the system is eligible is added. The reference formulas are given below [eq. 1-6]:

$$NPV = \sum_{t=0}^{N} \frac{(CI - CO)}{(1+r)^t} = DCI - DCO, \tag{1}$$

$$DCI = \sum_{t=1}^{N} \frac{\overline{\omega}_{self,c} \times E_{Out,t} \times p_t^c + \left[\left(1-\overline{\omega}_{self,c}\right) \times E_{Out,t} - E_{Ex,\,t}\right] \times p_t^s + \overline{\omega}_{self,c} \times E_{Out,t} \times \left(S_{REC}^u + C_{val,u}\right)}{(1+r)^t} + \sum_{t=1}^{N} \frac{R_{Ex,t} + E_{Ex,\,t} \times \left(S_{REC}^u + C_{val,u}\right) - S_{V,t}}{(1+r)^t} + \sum_{t=1}^{N_{TaxD}} \frac{\left(\frac{C_{inv}}{N_{TaxD}}\right) \times TaxD_u}{(1+r)^t}, \tag{2}$$

$$p_{t+1}^c = p_t^c \times (1 + inf_{el}), \tag{3}$$

$$p_{t+1}^s = p_t^s \times (1 + inf_{el}), \tag{4}$$

$$E_{Out,t} = t_r \times k_f \times \eta_m \times \eta_{bos} \times A_{cell} \times S, \tag{5}$$

$$E_{Out,t+1} = E_{Out,t} \times \left(1 - dE_f\right), \tag{6}$$

The energy supplied to consumers is equal, in each reference period, to the lower of the energy demand of the M consumers considered and the energy not self-consumed by prosumers [eq. 7-8]; it should be noted that the energy available for exchange each year considers the deterioration of the system.

$$E_{req} = \sum_{i=1}^{M} E_{req,i}, \tag{7}$$

$$E_{Ex,\,t} = \min\left[E_{req},\ \left(1 - \overline{\omega}_{self,c}\right) \times E_{Out,t}\right], \tag{8}$$

Revenue from sales to consumers is equal to the sum of the energy supplied to each of them and the respective prices [eq. 9].

$$R_{Ex,t} = \sum_{i=1}^{M} \left(E_{Exi,t} \times p_i^{REC}\right), \tag{9}$$

Where vulnerable households participate – requiring a portion of the incentives as additional income to lift them out of energy poverty once they join the REC, in accordance with the procedures set out in the new benefit distribution model to be outlined below – this portion is deducted from the revenue generated by prosumers [eq. 10-11].

$$V_t = \sum_{v=1}^{P} \left(E_{Exv,t} \times p_0^s + G_{v,t} - 10\% \times R_v\right), \tag{10}$$

$$S_{V,t} = \min\left[E_{Ex,\,t} \times \left(S_{\mathrm{REC}}^{\mathrm{u}} + C_{val,u}\right),\ V_t\right], \tag{11}$$

When determining the amount required by consumers, accounts are also taken of any expenditure related to the remaining energy that cannot be

supplied by prosumers and is obtained from the grid at market price [eq. 12].

$$G_{v,t} = \left(E_{req,v} - E_{Exv,t}\right) \times p^{c}_{t,v}, \quad (12)$$

When analyzing costs, account must be taken of expenditure relating to maintenance, insurance, taxes, electricity connection and inverter replacement, as well as the investment cost and expenses linked to the form of capital repayment chosen for the initiative [eq. 13].

$$DCO = \sum_{t=0}^{N_{debt}-1} \frac{\frac{C_{inv}}{N_{debt}} + \left(C_{inv} - C_{lcs,t}\right) \times r_d}{(1+r)^t} + \sum_{t=1}^{N} \frac{P_{Cm} \times C_{inv} \times (1+inf)^{t-1} + P_{Cass} \times C_{inv} \times (1+inf)^{t-1} + Sp_{el,t} \times P_{Ctax} + C_{i,t}}{(1+r)^t} + C_{ae}, \quad (13)$$

To account for cases where the inverter is replaced several times during the initiative's useful life, the associated cost has been modelled as follows [eq. 14].

$$C_{i,t} = \begin{cases} P_{Ci} \times C_{inv} \text{ if } t \in \{N^*, 2N^*, 3N^*, \ldots\} \text{ and } t < N \\ 0 \text{ otherwise} \end{cases}, \quad (14)$$

Sales relating to energy sold locally to consumers ($Sl_{el,t}$) are considered taxable, in the same way as sales to the GSE, indicated below as $Sr_{el,t}$ [eq. 15].

$$Sp_{el,t} = Sr_{el,t} + Sl_{el,t} = \left[\left(1 - \overline{\omega}_{self,c}\right) \times E_{Out,t} - E_{Ex,t}\right] \times p^{s}_{t} + R_{Ex,t}, \quad (15)$$

3.2. The software

To provide a quantitative assessment of the economic, environmental and social benefits, a dedicated software program written in Python, called S-REC, has been developed. This enables the economic and financial feasibility of the initiative to be assessed, alongside appropriate sensitivity analyses, scenario and risk analyses, and to study the distribution of the benefits obtained among the various REC members.

Using a dynamic and iterative approach, the software configures the structure of the REC in terms of energy production, investments, local energy sharing, pricing, and benefit distribution, whilst providing support where necessary to economically vulnerable households.

The software allows users to freely adjust the purchase prices, the sale prices and the exchange prices. It also suggests the most suitable prices for consumers identified as being in a vulnerable situation, which – together with the transfer of part of the incentives accrued on the energy exchanged with prosumers, in the most serious cases – ensures that they no longer suffer from energy poverty by participating in the REC.

More specifically, consumers can be characterized according to annual energy requirements, consumer type and, where applicable, income, market purchase price, and REC purchase price. A specific function identifies residential consumers experiencing energy poverty, defined as households whose energy expenditure exceeds 10 % of taxable income. This allows the simulation of scenarios with differentiated tariffs based on consumer type and income.

The software allows different combinations of parameters to identify the most economically sustainable REC structure. Consumers are served according to a FIFO (First In First Out) approach until potential saturation is reached. The process is dynamic, starting from a basic REC configuration and progressively including consumers, producers, and additional prosumers to meet residual demand and maximize overall self-consumption. The software also allows the evaluation of neighboring producers and the sizing of additional PV systems to cover local needs and achieve economies of scale.

Finally, it is worth highlighting the main differences between the developed software and others available on the market [49–52]. The software developed integrates economic analysis with the social dimension, offering users extensive customization options for their analyses. Unlike other software, it is possible to take into consideration different categories of consumers, including those experiencing energy poverty. It allows for the testing of different benefit-sharing methods with specific redistributive aims and the structuring of differentiated internal tariffs based on consumer type, market energy price, and income. It also enables assessment of the economic and social effects arising from the structure and benefit-sharing model under consideration, providing a clear picture of the benefits received by each individual participant. At the same time, it assesses the financial feasibility of this type of initiative, allowing the exploration of alternative configurations through sensitivity analyses, scenario and risk analyses, implemented to move beyond a deterministic approach and support the decision-making process. Furthermore, it can quantify the environmental benefit, in terms of tonnes of CO_2 equivalent avoided each year by installing PV systems within the REC.

Future developments include the integration of storage systems. In this study, a simplified case involving four residential prosumers and three consumers, two of whom are vulnerable, is used to illustrate the proposed benefit-sharing model.

3.3. Model for distributing benefits within the REC

Various methods for dividing revenues among prosumers are proposed in the literature: in some analyses, they are distributed uniformly, or based on the specific or partial consumption profiles of each member (i.e., they include an exchange price) [22]; in other approaches, greater weight is given to more virtuous behaviours in terms of the proposed exchange price and to better alignment between energy production and demand [11] and a bonus is then provided for lower-income participants [15]. In this context, the benefits have gone to prosumers, consumers, and households experiencing energy poverty. In contrast to these approaches, a mechanism is proposed that allocates benefits according to stakeholder categories, with each category receiving a percentage share of the total benefits, which are then distributed among the members of that category [18]. There is no single, predefined system for distributing benefits that has prevailed over others: depending on the political and geographical context (from the absence or presence of government incentives to the type of members constituting the REC) and the intentions of the stakeholders themselves, certain forms of distribution may prove more suitable than others.

This study proposes a new model that also takes neighbouring consumers into account, particularly households experiencing energy poverty. Prosumers who self-consume more gain greater benefits, due to savings on their bills and contributions linked to self-consumed energy, even though the share of sellable energy decreases, as the latter is valued at a lower price. Therefore, prosumers who adopt virtuous behaviour (in the sense of synchronising their consumption habits with PV production, i.e. during the day) are rewarded. At the same time, prosumers who sell part of their surplus to consumers gain additional income from the sale, which may exceed what they would receive from selling to the GSE in the case of consumers not in severe conditions of vulnerability, and from incentives, which are also granted for locally shared energy. Below, 'c' denotes consumers, whilst 'v' refers to consumers recognised as being in

severe energy poverty and who, therefore, also require incentives.

Firstly, the energy produced by the PV system is divided equally among the prosumers or Renewable Self-Consumers (RSCs) under consideration, by dividing the energy produced by the number of prosumers [eq. 16], in accordance with the principles of equitable treatment sought to ensure the satisfaction of RSCs and the long-term survival of the REC.

$$E_j = \frac{E_{Out,\ t}}{n_{RSC}} \text{ with } j = 1, \ldots, n_{RSC}, \tag{16}$$

The share of energy self-consumed by each prosumer is calculated as the product of the energy available to them and the specific self-consumption percentage of each RSC [eq. 17].

$$E_{SCj} = E_j \times \omega_{self\,j}, \tag{17}$$

The energy not self-consumed by each prosumer, which is to be sold to neighbouring consumers and/or the GSE, is equal to the difference between the energy produced at their disposal and the amount of energy self-consumed [eq. 18].

$$E_{NSCj} = E_j - E_{SCj}, \tag{18}$$

With a view to distributing the benefits associated with the sale of energy to consumers and the incentives accrued thereon evenly, consumers are served by all RSCs in equal parts until saturation. In this way, it is believed that we can prevent prosumers who would have obtained lower revenues from leaving the REC – it should be noted that membership is voluntary and non-binding – and avoid collusive behaviour, ensuring that sales are not concentrated in the hands of a few prosumers. Consequently, the energy supplied by the j-th prosumer to the i-th consumer is equal to the minimum of the surplus energy at their disposal and the energy required by the i-th consumer$(E_{C,\ i})$ divided by the number of prosumers considered [eq. 19].

$$E_{C,j\to i} = \min\left[E_{NSCj}, \frac{E_{C,\ i}}{n_{RSC}}\right], \tag{19}$$

The software implements the logic described above until the i-th consumer is satisfied and/or the energy available for exchange by the RSCs is exhausted ($\sum_{j=1}^{n_{RSC}} E_{NSCj} = 0$): when a j-th RSC exhausts its share of non-self-consumed energy, the number of RSCs active for the exchange – now denoted by$n_{RSC,a}$ – decreases accordingly, as does the remaining energy demand ($E_{C,\ i,r} = E_{C,\ i} - \frac{\sum_{j=1}^{n_{RSC}} E_{C,j\to i}}{n_{RSC}}$); in subsequent iterations, for the remaining active prosumers, the non-self-consumed energy available for exchange is reduced to$E_{NSCj,r} = E_{NSCj} - \frac{\sum_{j=1}^{n_{RSC}} E_{C,j\to i}}{n_{RSC}}$. The cycle continues and the active RSCs share the residual energy demand.

Therefore, formula (eq. 19) should be revised as follows [eq. 20], where at the first iteration $E_{C,\ i,r} = E_{C,\ i}$ and$E_{NSCj,r} = E_{NSCj}$, and$n_{RSC,a}$ denotes the number of active RSCs, i.e. those with non-self-consumed energy other than zero and therefore available to be exchanged with the consumer in question.

$$E_{C,j\to i} = \min\left[E_{NSCj,r}, \frac{E_{C,\ i,r}}{n_{RSC,a}}\right], \tag{20}$$

For consumers following the first consumer entered, the energy available to each RSC is initially reduced by the amount already transferred to consumer 1. This approach was deemed appropriate to distribute the proceeds from sales to consumers as equitably as possible whilst preventing RSCs from relying too heavily on energy sales: in this way, savings from self-consumption remain the key factor distinguishing the economic benefits achievable by prosumers, which in turn determines their profitability. Thus, the incentive for RSCs to adopt virtuous consumption behaviour remains, aligning energy use with its production as closely as possible. The revenue items associated with the various energy quotas presented are set out below.

The savings on the prosumer's bill are equal to the product of the self-consumed energy and the purchase price of energy on the market [eq. 21].

$$S_{billj} = E_{SCj} \times p_t^c, \tag{21}$$

The payment that each RSC can receive for self-consumed energy is calculated as the product of the proportion of self-consumed energy and the sum of the value-added payment set by ARERA and the premium tariff [eq. 22]; in the Italian context, the latter depends on the size of the plant, its location and the hourly zonal price.

$$S_{SCj} = E_{SCj} \times \left(C_{val,u} + \mathrm{S}^{\mathrm{u}}_{\mathrm{REC}}\right), \tag{22}$$

Where consumers are present and based on the energy exchanged, further incentives are granted. Expression (eq. 22) is modified as follows [eq. 23]:

$$S_{SCj} = \left(E_{SCj} + \sum_{i=1}^{M} E_{C,\ j\to i}\right) \times \left(C_{val,u} + \mathrm{S}^{\mathrm{u}}_{\mathrm{REC}}\right), \tag{23}$$

Furthermore, the energy sold to consumers that the generic RSC can serve is valued at the price p_i^{REC}, set according to the type of consumer. $R_{C,\ j} = \sum_{i=1}^{M} E_{C,j\to i} \times p_i^{REC}$ with $i = 1,\ 2,\ \ldots,\ M$

According to the FIFO logic, in line with the incremental approach adopted, consumer 2 receives energy if and only if the energy requested by consumer 1 is less than the energy not self-consumed by the RSCs.

At this point, it is worth highlighting the price differentiation within the REC. Lower prices are provided for vulnerable consumers, adjusted to ensure that energy expenditure does not exceed 10 % of their income; at the very least, the exchange price may be equal to the sale price to the GSE. The maximum price that can be imposed on a vth vulnerable consumer is calculated as the ratio between 10 % of their income and their energy requirement [eq. 24].

$$p_{v,max}^{REC} = \frac{10\% \times R_v}{E_{C,v}}, \tag{24}$$

Note that '$p_{v,max}^{REC}$' is set equal to 'p_0^s' if the latter is lower, so as not to make the sale of energy to vulnerable consumers less favourable than to the GSE and to avoid any tensions with and between prosumers. If the energy available to the REC is insufficient to cover the vulnerable consumer's entire energy requirement, account is also taken of the expenditure (G_v) associated with the portion of energy that they continue to purchase from the grid at market price [eq. 25]. A cost that will continue to be paid by the consumer.

$$G_v = \left(E_{C,v} - \sum_{j=1}^{n_{RSC}} E_{C,j\to v}\right) \times p_{t,v}^c, \tag{25}$$

The expression of the maximum price (eq. 24) applicable to the vth vulnerable consumer is modified accordingly [eq. 26], considering both the share of energy purchased from the REC and the remainder sourced from the national grid.

$$p_{v,max}^{REC} = \frac{10\% \times R_v - G_v}{\sum_{j=1}^{N} E_{C,j\to v}}, \tag{26}$$

It follows that the remaining energy for each prosumer to feed into the grid is equal to the difference between the energy not self-consumed and the energy exchanged with the M neighbouring consumers considered [eq. 27].

$$E_{Gj} = E_{NSCj} - \sum_{i=1}^{M} E_{C,j\to\ i}, \tag{27}$$

In turn, the revenues from sales to the GSE are equal to the product of the surplus energy not exchanged with neighbouring consumers and the grid sale price [eq. 28].

$$R_{salej} = E_{Gj} \times p_t^s, \tag{28}$$

Finally, in the presence of consumers, the total revenue achievable by the j-th prosumer is given by the sum of the income sources introduced previously: savings from self-consumption, incentives accrued on self-consumed and locally shared energy, and sales to neighbouring consumers and the GSE [eq. 29].

$$B_j = S_{billj} + S_{SCj} + R_{C,\ j} + R_{salej}, \tag{29}$$

From the consumer's perspective, the benefit is given by the product of the energy supplied by prosumers and the difference between the market purchase price and the price within the REC [eq. 30–31].

$$E_{Ex,i} = \sum_{j=1}^{n_{RSC}} E_{C,j\rightarrow\ i}, \tag{30}$$

$$B_i = E_{Ex,i} \times \ \left(p_t^c - p_i^{REC}\right), \tag{31}$$

Where there are consumers in severe energy poverty, provision is made for a share of the incentives accrued from the exchange to be transferred to them; therefore, equation (eq. 23) is revised as follows [eq. 32].

$$S_{SCj} = \left(E_{SCj} + \sum_{i=1}^{M} E_{C,\ j\rightarrow i}\right) \ \times \left(C_{val,u} + \mathrm{S}_{\mathrm{REC}}^{\mathrm{u}}\right) - S_{j\rightarrow v}, \tag{32}$$

The share of state subsidies transferred to vulnerable consumers by each prosumer can be expressed as the sum of the ratio between the incentives that P consumers in severe energy poverty would require to escape that condition and the number of prosumers interacting with each of them. In this way, the transfer of benefits linked to the incentives is also distributed evenly among the prosumers [eq. 33].

$$S_{j\rightarrow v} = \sum_{v=1}^{P} \frac{S_v}{N_v}, \tag{33}$$

Obviously, the NPV for prosumers also changes: any incentives transferred to vulnerable households are subtracted from each year's figure, as shown in (eq. 2).

More specifically, the incentives transferred to the vth consumer in a state of severe vulnerability are calculated as follows. If $p_{v,max}^{REC} < p_0^s$, the exchange price for the consumer in question is set equal to the sale price to the GSE, so that the prosumer(s) supplying them with energy do not lose direct benefits; on the other hand, the incentives on the exchanged energy are shared with the severely vulnerable consumer to such an extent that they no longer suffer from energy poverty thanks to the extra income provided by state subsidies. The incentives now serve a dual purpose: on the one hand, they drive the uptake of renewables and configurations for widespread self-consumption; on the other, they act as a measure to combat energy poverty and avoid exacerbating social inequalities. This proposed new approach is supported by the fact that several studies have demonstrated the cost-effectiveness of such investments on the prosumer side, regardless of the presence of state incentives.

In practice, if a vulnerable consumer's energy expenditure remains above 10 % of their income whilst sourcing energy from the REC at the minimum price (p_0^s), they receive a share of the incentives accrued on the energy traded locally by the prosumer(s) with whom they interact, with an equal share allocated to each prosumer to ensure a uniform distribution of both costs and benefits [eq. 34-35].

$$E_{Ex,v} = \sum_{j=1}^{n_{RSC}} E_{C,j\rightarrow v}, \tag{34}$$

$$V_v = p_0^s \times E_{Ex,v} + G_v - 10\% \ \times R_v, \tag{35}$$

In extreme cases, they are granted the full amount of the incentives [eq. 36].

$$S_v = \min\ \left[E_{Ex,\ v} \times \left(\mathrm{S}_{\mathrm{REC}}^{\mathrm{u}} + C_{val,u}\right),\ V_v\ \right], \tag{36}$$

Therefore, for consumers in severely vulnerable circumstances, the expression of the total gain obtainable through the REC (eq. 31) is modified as follows [eq. 37], including the contribution from incentives as well as the savings on bills achieved by sourcing from the REC at a lower exchange price than the market price.

$$B_v = E_{Ex,v} \times \ \left(p_t^c - p_{v,\ max}^{REC}\right) + S_v, \tag{37}$$

In percentage terms, the total gain for each REC member is calculated as the ratio of the individual participant's total benefit to the sum of the benefits obtained by prosumers and consumers [eq. 38–40].

$$B_{tot} = \sum_{j=1}^{n_{RSC}} B_j + \sum_{i=1}^{M} B_i, \tag{38}$$

$$BD_j = \frac{B_j}{B_{tot}}, \tag{39}$$

$$BD_i = \frac{B_i}{B_{tot}}, \tag{40}$$

Finally, the NPV for each individual prosumer is calculated by multiplying the project's NPV by the percentage share of the benefits achievable from the j-th RSC, without considering the percentage contribution of consumers [eq. 41].

$$NPV_j = NPV \times \frac{BD_j}{\sum_{j=1}^{n_{RSC}} BD_j} \ with\ j = 1, \ldots, n_{RSC}, \tag{41}$$

In the presence of consumers, the prosumer-side NPV will increase due to the energy sold to consumers (and no longer to the grid) at a price higher than the tariff set by the GSE, and thanks to the higher incentives paid in the Italian context, including for locally traded energy. On the other hand, the contribution linked to the sale of surpluses to the GSE will decrease, albeit less than proportionally, compared to the case where consumers are absent.

Table 2
Input data.

Variable	Symbol	Value
Energy produced in the first year	$E_{Out,t}$	38,996.3 kWh
Average self-consumption	$\overline{\omega}_{self,c}$	50 %
Lifespan of the PV system	N	20 years
Opportunity cost of capital	r	5 %
Purchase price at time t	p_t^c	0.3 €/kWh
Selling price at time t	p_t^s	0.12 €/kWh
Premium tariff + ARERA fee	$S_{REC}^{u} + C_{val,u}$	0.124 €/kWh
Tax deduction period	N_{TaxD}	10 y
Initial investment per unit	$C_{inv,\ unit}$	2000 €/kW
Tax deduction	$TaxD_u$	36 %
Average annual insolation	t_r	1450 kWh/m² × a
Optimal angle coefficient	k_f	1.13
Module efficiency	η_m	20 %
System efficiency balance	η_{bos}	85 %
Active module area	A_{cell}	7 m²/kW
Size of the PV system	S	20 kW
Power reduction	dE_f	0.8 %
Electricity inflation rate	inf_{el}	3 %
Value added tax	VAT	10 %
Equity ratio	E	0 %
Capital repayment period	N_{debt}	10 years
Mortgage interest rate	r_d	3 %
Percentage maintenance cost	P_{Cm}	2 %
Eurozone inflation rate	inf	3 %
Insurance cost percentage	P_{Cass}	1 %
Taxes	P_{Ctax}	27 %
Percentage cost of inverter	P_{Ci}	15 %
Electricity connection cost	C_{ae}	4000 €
Inverter replacement	N*	10 years

3.4. Input data

Using the S-REC software, an economic analysis of the REC was carried out within the Italian regulatory framework; the model input data are presented in Table 2 and were selected in accordance with the literature [11,15,18,53–57].

The size of the PV system (20 kW) was selected to cover the aggregate annual electricity demand of the REC participants (approximately 31,498 kWh in the first year). Based on the assumed irradiation and system-efficiency parameters reported in Table 2, the system is expected to generate 38,996 kWh in the first year, ensuring sufficient production to meet local demand while accounting for annual PV degradation. This sizing approach, based on matching expected PV generation with aggregate electricity demand, is consistent with the methodology adopted in previous studies on RECs [11]. Furthermore, the useful life of the PV plant has been estimated at twenty years [18], during which the energy produced will decrease from 38,996.3 kWh in the first year to 33,476.9 kWh in the last, due to the annual loss of efficiency of PV systems.

The project is fully financed through debt, with a debt repayment period of ten years. The unit investment cost has been estimated at 2000 €/kW. This cost represents the installed cost of the PV system rather than the cost of PV modules alone. It includes the purchase of PV modules, inverters, mounting structures, electrical components, design and permitting costs, installation labor, grid connection costs, and other balance-of-system expenses.

This value is slightly more conservative than those adopted in some previous studies [11,15], reflecting a cautious assumption intended to account for potential variability in installation and connection costs. No storage systems have been included. Part of the surplus energy is sold to neighbouring consumers, based on their annual energy requirements; the remainder is sold to the grid under the conditions set by the GSE.

The price bands were not considered because a flat-rate tariff was applied. With the flat-rate tariff, you pay the same price for the energy component at any time of day. The fixed-rate power offer is ideal for anyone who also - or especially - need to consume energy during the day. The cost of electricity in the bill, estimated based on various residential utility bills and Eurostat data, is approximately 0.30 €/kWh and represents the savings achievable through self-consumption. This value is consistent with those adopted in other studies [15]. However, comparable values have also been reported in other contributions [18], and, given the variability of retail electricity prices over time, alternative low- and high-market-price scenarios can be considered, as suggested in the literature [11,22]. The electricity tariff includes energy supply costs and the variable part of system charges, associated with every kWh that is no longer purchased from the grid.

The energy fed into the grid is instead remunerated at an average price of 0.12 €/kWh, according to GSE data and previous studies [15,18].

For energy self-consumed within an energy community, the GSE provides an incentive tariff of 0.124 €/kWh. This consists of a fixed component of 0.08 €/kWh and a variable component calculated as min (0.04; max(0; 0.180 – hourly zonal price)) €/kWh, with an overall cap of 0.120 €/kWh. An additional incentive of 0.004 €/kWh is also granted for photovoltaic systems located in the regions of Central Italy. This value is consistent with those adopted in the literature, although it represents an optimistic assumption, as the maximum value within the eligible range was considered. In other cases, more conservative assumptions corresponding to the minimum value of the range or to an intermediate value could be adopted [11,15,18,22].

Four residential prosumers and three domestic consumers have been considered; it should be noted that two of them are in a state of energy poverty, with varying degrees of vulnerability.

4. Results

This section demonstrates the effectiveness of the benefit distribution model presented through the following case study. The initiative analysed involves the installation of a 20 kW PV system by four residential prosumers (RSCs).

The monocrystalline PV panels are located on the roofs of their homes. Consequently, the prosumers do not incur any additional costs associated with the purchase or rental of property on which to install the system. The consumption profiles of the assumed prosumers are 80 %, 65 %, 35 % and 20 % for RSC1, RSC2, RSC3 and RSC4 respectively, resulting in an average self-consumption rate of 50 %. The average self-consumption rate of 50 % was selected in accordance with values commonly adopted in the literature for residential PV systems operating without battery storage [11,15,18]. This assumption reflects a realistic level of on-site electricity use under typical household load profiles. To represent heterogeneity among participants, four prosumers were assigned different self-consumption rates (80 %, 65 %, 35 %, and 20 %), covering a broad range of behavioral patterns and load synchronization with PV generation. Higher values represent households that shift a larger share of their consumption to daylight hours, while lower values reflect less favorable consumption profiles. The average of these four values corresponds to the overall self-consumption rate of 50 % assumed for the REC.

Neighbouring consumers are characterised as follows (Table 3); the purchase price for domestic users is assumed to be 0.3 €/kWh, in line with Table 2.

Residential consumers are aware of the benefits of joining a REC, but do not have the financial means to invest in the system. Given that with appropriate prices, lower than the market price for consumers and higher than the selling price to the GSE for prosumers, a win-win situation is achieved for both parties, prosumers supply surplus energy to consumers at an intermediate (exchange) price.

The intermediate exchange price is derived from annual average values following a logic validated by the literature [15,22]. Specifically, it is calculated as the arithmetic mean between the average annual retail electricity purchase price for residential consumers (0.30 €/kWh) and the average annual selling price to the GSE (0.12 €/kWh), resulting in an exchange price of 0.21 €/kWh. Monthly variations in electricity prices are not considered, as the analysis assumes flat tariffs and focuses on annual average values, in line with the long-term discounted cash flow approach adopted in the study.

In turn, the two domestic consumers in energy poverty are not served under the same conditions as the non-vulnerable residential consumer. The price is chosen as follows: if the maximum price to be applied to ensure they no longer fall into energy poverty (i.e. their energy expenditure no longer exceeds 10 % of their income) is greater than or equal to the selling price to the GSE, it is likely to be deemed acceptable by prosumers and is set as such; otherwise, for prices lower than the selling price to the GSE, the tariff is set equal to the selling price and the incentives on the energy exchanged with prosumers are transferred to the vulnerable consumer, to the extent that they act as additional income and ensure that they are no longer in energy poverty. The transfer of incentives can serve as a further lever to overcome resistance to change and the NIMBY (Not In My Back Yard) syndrome, which refers to residents' opposition to projects perceived as undesirable when located near their homes. In this context, direct economic benefits can increase local acceptance and encourage participation in the REC. The resulting differentiated prices are presented in Table 4.

As regards vulnerable consumer 1, characterised by an income of

Table 3
Characterization of neighboring consumers.

Type	Energy demand [kWh/year]	Market price [€/kWh]	Income [€/year]
Vulnerable 1	3000	0.3	4000
Vulnerable 2	3000	0.3	3000
Residential	6000	0.3	30,000

Table 4
Differentiated REC prices to account for vulnerabilities.

Type	Energy demand [kWh/year]	Market price [€/kWh]	Income [€/year]	REC price [€/kWh]	Receives incentives
Vulnerable 1	3000	0.3	4000	0.133	No
Vulnerable 2	3000	0.3	3000	0.12	Yes
Residential	6000	0.3	30,000	0.21	No

4000 €, an energy demand of 3000 kWh and a market price of 0.3 €/kWh, it is sufficient to apply a lower price of 0.133 €/kWh to lift them out of their vulnerable status, in accordance with eq. (24); this situation is also profitable for prosumers, who still earn more than they would by selling to the grid. Given that the price required to ensure that the energy expenditure of vulnerable consumer 2 no longer exceeds 10 % of their income would be lower than the selling price to the GSE (0.10 €/kWh vs 0.12 €/kWh) and prosumers might be reluctant to make such concessions, a price in line with the selling price to the GSE is chosen, so that the RSCs do not have lower revenues than in the case where vulnerable consumers are not included in the configuration, thereby encouraging its acceptance. On the other hand, the consumer in question receives, in part or in full, the incentives accrued on the shared energy supplied to them by prosumers: additional benefits which, it should be emphasised, the RSCs would not have received in their absence, given that the surplus energy would have been sold to the GSE and would therefore not have been eligible for incentives. Furthermore, given that vulnerable consumer 2 does not require the full amount of revenue linked to state subsidies on self-consumed energy, the remaining portion is allocated in equal parts to the RSCs from which they are supplied. Table 5 shows the results relating to the distribution of benefits.

Each prosumer has five sources of income, the size of which varies depending on the respective percentages of energy self-consumed, exchanged with neighbouring consumers and sold to the GSE. For example, RSC 2, which self-consumes 65 % of the energy at its disposal, equivalent to 6366.9 kWh, achieves savings on its electricity bill amounting to 1901.07 €, based on a market price of 0.3 €/kWh; furthermore, it receives self-consumption incentives totalling 785.77 €, calculated by multiplying the 6366.9 kWh of self-consumed energy by the applicable unit incentive (0.124 €/kWh).

The remaining 35 %, corresponding to the portion of energy not self-consumed, amounting to 3412.18 kWh, is sold partly to consumers and partly to the grid. More specifically, 3350.06 kWh are exchanged by RSC 2 with consumers; notice that RSCs 3 and 4 also achieve exchanges of the same magnitude, whilst RSC 1 shares 1949.82 kWh, reaching saturation: this demonstrates compliance with the logic outlined previously. RSC 2 earns a further 578.26 € from sales to consumers and 415.41 € from incentives for locally shared energy, within the Italian context; from the latter, the share of incentives to be transferred to vulnerable consumer C2 must be deducted, amounting to 15 € and equal for all prosumers trading with the consumer in question. This confirms the logic followed in seeking to distribute costs and benefits as evenly as possible among the RSCs. The remaining energy (62.12 kWh) is sold to the grid at a price of 0.12 €/kWh, yielding 7.45 €. Adding up the revenue items gives the total profit for prosumer 2: 3672.97 €. Finally, the total profit is divided by the sum of the benefits accrued by the prosumers and consumers in question, amounting to 14,255 €: this results in a benefit distribution of 25.77 % for RSC2. Comparing the values of the total benefits achieved without consumers (3096.31 € in the case of RSC2), it can be seen that the presence of consumers, whether vulnerable or not, brings greater benefits for all prosumers; furthermore, RSC2, RSC3 and RSC4, which self-consume less than RSC1, bridge part of the gap with prosumer 1, due to the larger-scale exchanges they have with neighbouring consumers, thanks to prices that are not lower than the selling price and the provision of incentives for locally exchanged energy. Table 6 presents the benefits achievable by the various consumers.

Consumer C1 purchases 3000 kWh from the REC and is supplied in equal parts by the four prosumers. If C1 previously had an energy bill of 900 €, calculated as the product of the 3000 kWh required and the market price for residential users (0.3 €/kWh), by joining the REC their energy bill is reduced to 399 €: each RSC supplies 750 kWh at the lower price of 0.133 €/kWh, which is more suited to their vulnerable status. This results in a saving on the bill of 501 €, calculated as the difference between energy expenditure with and without the REC. On the other hand, prosumers achieve higher revenues compared to the scenario without consumer C1, due to the incentives and the greater earnings derived from the exchange compared to selling to the GSE at a price of 0.12 €/kWh.

In turn, consumer C2 saves 540 € on their bill, calculated as the difference between purchasing 3000 kWh at the market price of 0.3 €/kWh – common to the three residential consumers considered – and receiving their supply from the REC at the lower price of 0.12 €/kWh. Furthermore, given their severe vulnerability, they require a portion of the incentives on the traded energy to be allocated to them, totalling 60 €, given an energy requirement of 3000 kWh, 10 % of his income, which corresponds to 300 €, and an applicable price of 0.12 €/kWh, instead of the price of 0.1 €/kWh, which would not likely be accepted by the prosumers as it is lower than the selling price to the grid. The incentives required are allocated in equal parts by the four prosumers from whom he sources his energy, amounting to 15 € per RSC, as shown previously

Table 5
Revenues divided according to consumption profile and support for vulnerable consumers.

	RSC1	RSC2	RSC3	RSC4
Percentage of self-consumption	80 %	65 %	35 %	20 %
Available energy [kWh]	9749.075	9749.075	9749.075	9749.075
Self-consumed energy [kWh]	7799.26	6336.9	3412.18	1949.82
Non-self-consumed energy [kWh]	1949.82	3412.18	6336.9	7799.26
Energy sold to consumers [kWh]	1949.82	3350.06	3350.06	3350.06
Residual energy [kWh]	0	62.12	2986.84	4449.2
Incentives for self-consumption [€]	967.11	785.77	423.11	241.78
Incentives for shared energy [€]	241.78	415.41	415.41	415.41
Incentives transferred to vulnerable consumers [€]	15	15	15	15
Total incentives accrued [€]	1193.89	1186.18	823.52	642.19
Savings on bills [€]	2339.78	1901.07	1023.65	584.95
Profit from sales to consumers [€]	284.21	578.26	578.26	578.26
Profit from sale to the GSE [€]	0	7.45	358.42	533.9
Total profit [€]	3817.88	3672.97	2783.85	2339.3
Percentage distribution of benefits	26.78 %	25.77 %	19.53 %	16.41 %
Percentage expenditure savings	16.41 %	13.34 %	7.18 %	4.10 %
Total benefit excluding consumers [€]	3540.97	3096.31	2207.19	1762.64

Table 6
Benefits achieved by consumers.

	C1	C2	C3
Type	Vulnerable	Vulnerable	Residential
Energy purchased from RSCs [kWh]	3000	3000	6000
Pre-REC energy expenditure [€]	900	900	1800
Post-REC energy expenditure [€]	399	360	1260
Savings on energy bills [€]	501	540	540
Incentives received [€]	0	60	0
Total benefit [€]	501	600	540
Percentage distribution of benefits	3.51 %	4.21 %	3.79 %

in Table 5. Consequently, the total benefit achievable by consumer C2 is 600 €, given by the sum of the savings on the bill (540 €) and the incentives granted to him (60 €). Dividing consumer C2's total gain by the sum of the benefits received by prosumers and consumers, amounting to 14,255 €, yields a benefit distribution of 4.21 % for consumer C2. This is higher than the benefit achieved even by non-vulnerable residential consumer C3, who has a total benefit of 540 €, demonstrating the effectiveness of the new form of redistribution envisaged. It can be seen that, thanks to more affordable prices that take vulnerability into account (0.133 €/kWh and 0.12 €/kWh) and the additional income listed as 'Incentives received [€]' (amounting to 60 €), vulnerable consumers are no longer in a state of energy poverty: their energy expenditure now does not exceed 10 % of their income. Consumer C1 has energy expenditure of 399 €: with an income of 4000 €, 10 % amounts to 400 €.

In Figs. 1 and 2, the results show the distribution of benefits among REC participants in both percentage and absolute terms. The blue bars represent the benefits arising from reduced bills, the pink and grey bars represent the benefits arising from the sale of energy to the REC and the grid respectively, and the yellow bars represent the benefits arising from incentives. This visualisation allows us to ascertain how the new allocation of incentives (the yellow bar for consumer C2) makes it possible to effectively combat energy poverty.

Prosumers still earn higher revenues than in the scenario where there are no consumers, whether vulnerable or not, by selling their energy surpluses to them; consumers, on the other hand, achieve savings on their bills by purchasing energy from the REC at stable prices that are lower than market rates. Under the new distribution method, C1 and C2 consumers are no longer in energy poverty, thanks to prices that take account of their vulnerable status and the transfer of part of the incentives on the energy traded. Furthermore, the presence of consumers promotes a more balanced distribution of benefits among prosumers and helps retain RSCs that would otherwise receive lower returns due to lower self-consumption rates. This is crucial to ensuring the long-term viability of the REC. Finally, it was considered of interest to examine how the main economic indicators vary depending on the presence of consumers and incentives (Table 7).

The highest NPV value (101,287 €) is achieved when incentives are combined with vulnerable consumers: this highlights the importance of incentives in encouraging local energy exchange, with consumers of all types, even when the exchange price is lower for certain categories of users. A win-win situation for prosumers and consumers is confirmed: energy sharing leads to economic benefits for both parties, particularly for prosumers who have invested in the project, as their investment would be recouped sooner (2 years vs 5 years).

The pre-discounted cash flow trends generated by the software are presented in Figs. 3 and 4 for two representative scenarios. Fig. 3 illustrates the annual costs and revenues in the absence of consumers and incentives, while Fig. 4 presents the corresponding cash flow trends for the best-case scenario, which includes both consumers (vulnerable and non-vulnerable) and incentives. Together, these figures confirm the economic feasibility of local energy sharing and highlight the higher revenues achievable when surplus electricity is sold to consumers and incentives are available.

Although costs increase slightly compared with the case shown in Fig. 3, due to taxes on sales to consumers, revenues increase more than proportionally owing to local energy sales and the incentives received, net of the share transferred to vulnerable households. Consequently, the combination of consumers and incentives for shared energy yields the best performance in terms of the main economic indicators, while also generating significant social and environmental benefits through the local sharing of clean energy at more affordable prices.

5. Discussion

The results obtained highlight how RECs can serve as an effective tool not only for the energy transition but also for promoting social and territorial equity. In line with the literature, RECs are decentralized energy models capable of integrating environmental, economic, and social benefits, contributing to the achievement of sustainable development goals, particularly access to clean energy and the creation of sustainable cities [11,58,59]. However, the analysed contribution introduces an innovative element: the possibility of implementing targeted redistributive mechanisms that allow for a concrete approach to addressing the problem of energy poverty.

From an economic perspective, the results confirm what has been widely highlighted in the literature, namely that the profitability of RECs depends significantly on the level of self-consumption and the presence of incentives [15,18]. However, another significant aspect emerges: the inclusion of consumers, including vulnerable ones, and the local exchange of energy contribute to increasing the overall profitability of the system. This result reinforces the idea that RECs can create a win-win situation between prosumers and consumers, improving revenue streams through the local utilization of energy rather than selling it to the grid [60,61]. In this sense, the model moves beyond a traditional view in which social equity and economic efficiency are seen as potentially conflicting.

A further element of consistency with the literature concerns the central role of benefit distribution. Previous studies emphasize that the

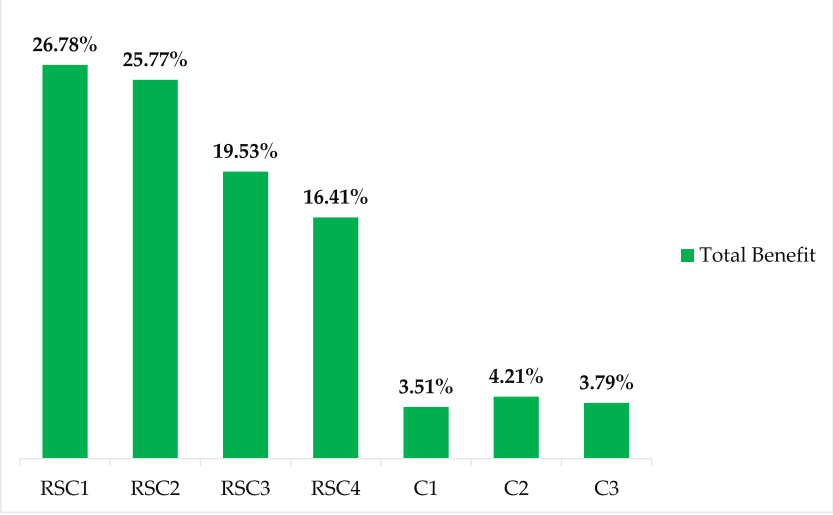


Fig. 1. Distribution of total benefits between prosumers and consumers (%).

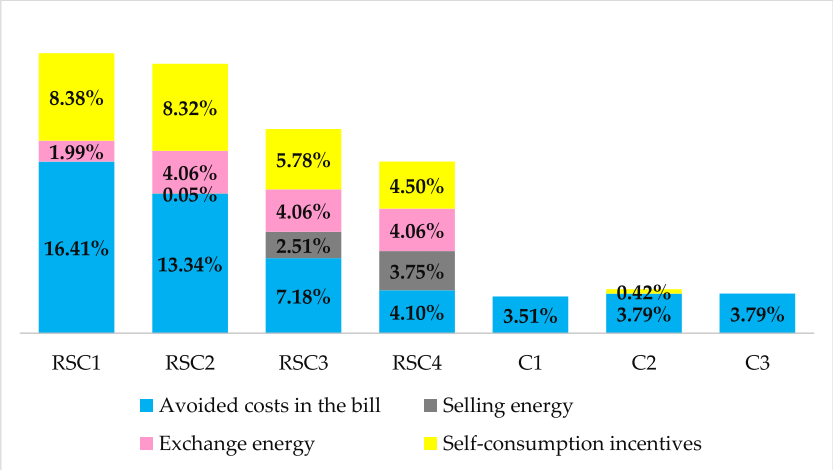


Fig. 2. Distribution of benefits between prosumers and consumers by type (%).

Table 7
Economic indicator results.

	NPV [€]	NPV/Size [€/kW]	PI	IRR [%]	DPBT [years]
Excluding consumers and incentives	53,599	2680	2.22	32.08	5
Excluding consumers and including incentives	81,906	4095	2.86	52.75	3
Including consumers and excluding incentives	55,185	2759	2.25	36.14	4
Including consumer and incentive	101,287	5064	3.30	70.00	2

sustainability of RECs depends on the ability to balance individual and collective interests, favouring fair and inclusive distribution models [33,34]. The results obtained confirm this evidence, showing how redistributive mechanisms, based on differentiated pricing and incentive transfers, can significantly improve equity without compromising overall profitability. Furthermore, in line with the preferences expressed by citizens, there is a strong inclination toward models that combine individual benefits and solidarity, with a willingness to forgo a portion of one's own benefits to support the most vulnerable groups [15].

Particularly relevant is the contribution on the topic of energy poverty. In this context, the literature highlights how RECs also adopt operational measures such as financial contributions, energy vouchers, tariff reductions, support for accessing public subsidies, and energy efficiency education initiatives, in addition to building interventions to reduce consumption [19,62]. These practices reinforce the potential of RECs as concrete tools to combat energy poverty. The literature highlights how RECs can serve as a useful tool to improve access to energy and reduce inequalities [17,63]. The distribution models proposed in this study demonstrate support for vulnerable consumers to alleviate energy poverty. However, these aspects must be validated through real-world social analyses to demonstrate how RECs can effectively be designed as tools for energy welfare, as well as for ecological transition.

Unlike models that assume a uniform distribution of benefits among vulnerable users, resulting in a reduction of individual benefits as the number of beneficiaries increases [15,18], the proposed model introduces a detailed characterization of consumers (income, energy expenditure), enabling more targeted and equitable interventions.

Finally, the results also highlight significant implications for public

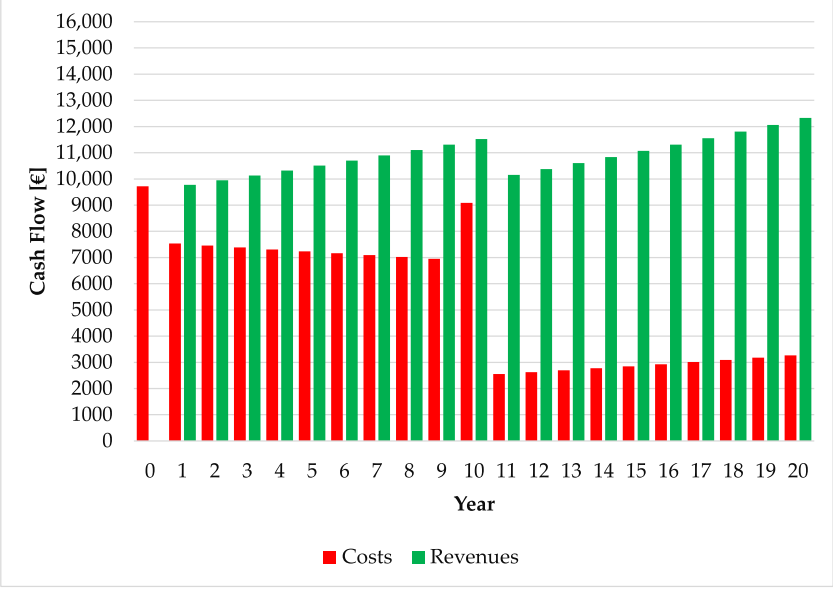


Fig. 3. Cash flows in the absence of consumers and incentives.

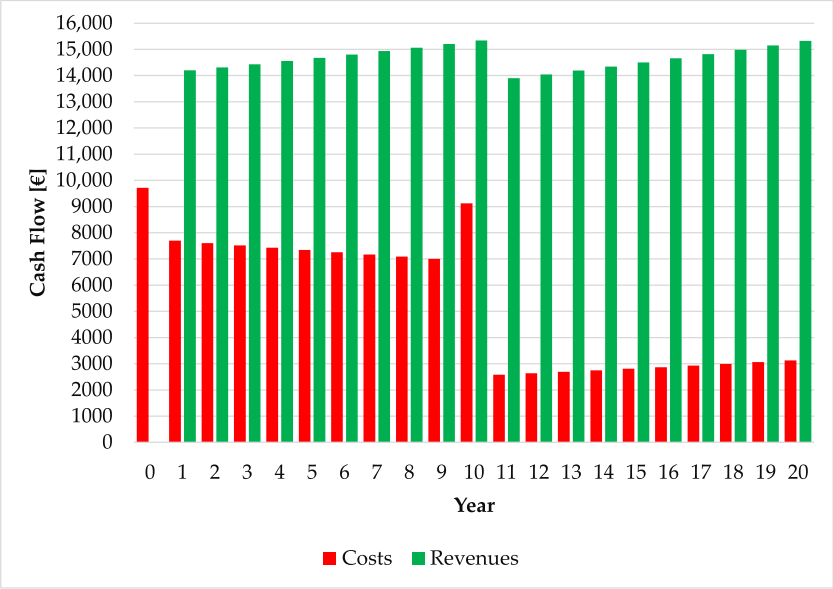


Fig. 4. Cash flows with consumers and incentives.

policy. In line with what is suggested in the literature, there is a need for a stable regulatory framework and appropriate incentive tools to promote the diffusion of RECs [31,64]. However, the proposed model also suggests an evolution in the role of incentives, which could be used not only to promote renewable energy but also to support social equity objectives. This approach is consistent with the broader objective of ensuring that the energy transition is accompanied by social inclusion, participation, and equity.

More broadly, the proposed model can serve as an ex-ante decision-support tool within the energy nexus, helping policymakers and local stakeholders design Renewable Energy Communities that simultaneously enhance economic viability, social inclusion, and the sustainable management of local energy resources.

6. Conclusions

This study examined the role of RECs as tools capable of combining economic efficiency and social equity, with a particular focus on the challenge of energy poverty. A new benefit-sharing model was developed, based on differentiated tariffs and the targeted allocation of incentives, and implemented using a dedicated simulation tool (S-REC), which enables the analysis of energy, economic, and social dynamics within different REC configurations.

The results highlight that the energy transition should not be accessible only to the wealthiest actors. In this context, the proposed model enables a more equitable distribution of benefits, allowing RECs to act as effective tools for redistributing value and improving access to reliable and sustainable energy for vulnerable groups. In particular, vulnerable consumers can obtain sufficient economic benefits to overcome energy poverty through an appropriate redistribution of incentives associated with locally shared energy, while ensuring that prosumers receive adequate remuneration, thereby preserving their participation and the profitability of investments. The results confirm that a more equitable distribution mechanism can support vulnerable users without negatively affecting prosumers.

Furthermore, although prosumers with higher self-consumption continue to earn higher individual returns, those with lower self-consumption significantly increase their revenues through the sale of energy to consumers and incentives related to locally shared energy. While the profitability of REC initiatives remains strongly influenced by self-consumption rates and electricity prices, the results show that the presence of both incentives and consumers, including vulnerable ones, leads to more favourable economic outcomes. These results reinforce the role of RECs in supporting sustainable development and contributing to the achievement of SDG 7 by ensuring access to affordable, reliable, and sustainable energy.

At the same time, the effectiveness of such models depends on adequate public support. The results suggest that incentive programs should evolve to better integrate social welfare objectives, particularly in mature photovoltaic markets where further reductions in trading prices may not be acceptable to prosumers. In this regard, integrating social criteria into incentive allocation mechanisms, by rewarding energy sharing with vulnerable users, represents a key policy direction for improving both equity and system performance.

However, this study also has some limitations. The analysis is based on a simulated case study within a specific regulatory context, and therefore further empirical validation in real-world settings is needed. Future research could extend the model by incorporating additional technological components, such as energy storage systems, and by exploring its applicability in different regulatory and socio-economic contexts.

Finally, the implications for policy are significant. Supporting the deployment of RECs through stable regulatory frameworks and targeted incentives can foster both the adoption of renewable energy and social inclusion, strengthening the link between technological innovation and equitable sustainability.

CRediT authorship contribution statement

Idiano D'Adamo: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Lorenzo D'Agostino:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Massimo Gastaldi:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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